

# Bill shape distinguishes Antarctic penguin species

The Day-1 result — one source, a branded Typst PDF

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## 1 Introduction

The base-R `penguins` dataset records body measurements for 342 penguins of three species collected at Palmer Station, Antarctica (Gorman et al., 2014). It reached the R community first through the `palmerpenguins` package (Horst et al., 2020). Here we ask a simple question: **does bill shape tell the species apart?** We render the answer as a branded PDF with **Typst**, a modern typesetting system that ships inside Quarto, so there is no LaTeX involved.

## 2 Species at a glance

Table 1

### Penguins of the Palmer Archipelago

Mean measurements per species

| Species   | N   | Bill (mm) |       | Body mass (g) |
|-----------|-----|-----------|-------|---------------|
|           |     | Length    | Depth |               |
| Adelie    | 151 | 38.8      | 18.3  | 3 701         |
| Gentoo    | 123 | 47.5      | 15.0  | 5 076         |
| Chinstrap | 68  | 48.8      | 18.4  | 3 733         |

Gentoo is the heaviest on average, at 5 076 g, the `large-bodied` species of the three.

### 3 Bill length versus depth

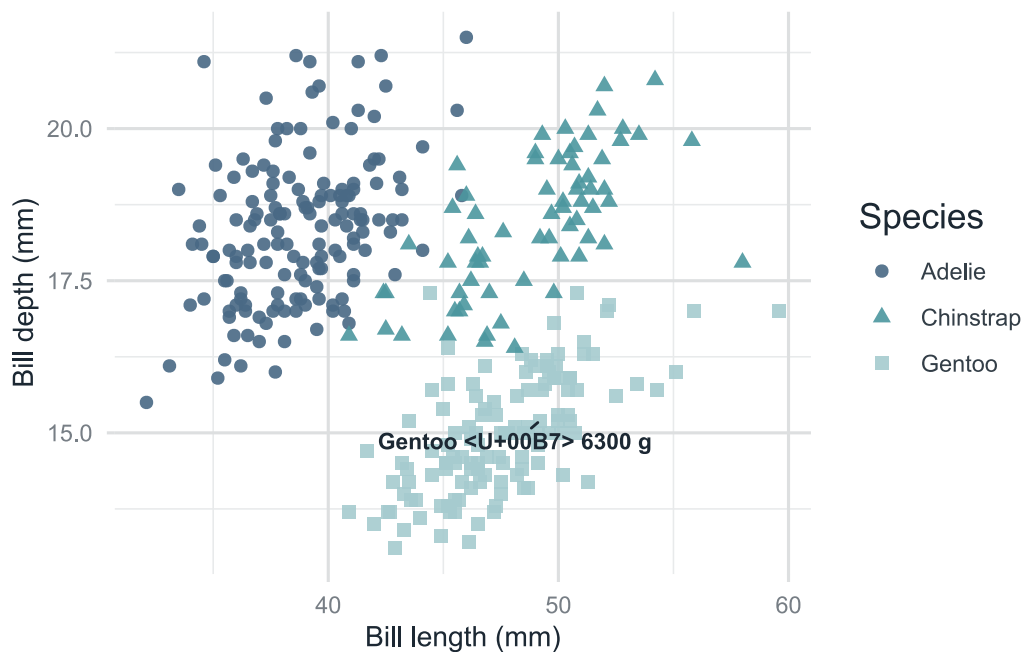


Figure 1: Bill length versus depth by species. The single heaviest penguin is labeled.

The three species separate cleanly in bill shape. This is the manuscript-ready version of the same figure from the running HTML report, now in a branded PDF.

### 4 Conclusion

One `.qmd`, one `_brand.yml`, and `format: typst` give a typeset document with citations, and carry the `RaukR` palette through layout, tables, and figures.

### References

- Gorman, K. B., Williams, T. D., & Fraser, W. R. (2014). Ecological sexual dimorphism and environmental variability within a community of Antarctic penguins (genus *Pygoscelis*). *PLoS ONE*, 9(3), e90081. <https://doi.org/10.1371/journal.pone.0090081>
- Horst, A. M., Hill, A. P., & Gorman, K. B. (2020). *palmerpenguins*: Palmer Archipelago (Antarctica) penguin data. <https://doi.org/10.5281/zenodo.3960218>